

ENPH 453 Advanced Physics Laboratory
Group 2 Letter of Intent 2

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Scientific Motivation & Theoretical Prediction

Electron Spin Resonance (ESR) probes systems with unpaired electron spins by measuring resonant absorption of microwave radiation in the presence of a static magnetic field [1]. For an electron with $S = \frac{1}{2}$, a magnetic field H_0 lifts the degeneracy of the spin states through the Zeeman interaction (Eq. 1)

$$\widehat{\mathcal{H}_z} = g\mu_B H_0 \widehat{S_z} \quad (1)$$

where g is the Landé g-value, $\widehat{\mathcal{H}_s}$ is the spin Hamiltonian, and μ_B is the Bohr magneton (9.274×10^{-28} JG^{-1}) [2]. This interaction splits the two spin states ($m_s = \pm\frac{1}{2}$) by an energy $\Delta E = g\mu_B H_0$ [3]. Resonance occurs when the microwave photon energy matches this splitting, when $h\nu = g\mu_B H_0$ [3].

In this experiment, the microwave frequency ν is held fixed while the magnetic field is swept, so resonance is observed at a specific H_0, allowing for the determination of g [4]. Diphenyl Picryl Hidrazyl (DPPH) is used for calibration, producing one narrow resonance line with $g = 2.0036$ [5].

MgO crystals contain transition-metal impurities (Fe^{3+}, Cr^{3+}, Mn^{2+}), with multiple spin states that may interact with nearby nuclei [6], [7], [8]. As a result, their ESR spectra can exhibit multiple resonance lines: hyperfine splitting from electron-nuclear spin coupling gives equally spaced components, while fine (crystal field) interactions due to cubic site symmetry can introduce angular dependence [3]. Thus, sweeping the magnetic field should yield a single sharp resonance from DPPH as well as structured multiplets from MgO impurities from which the ions and magnetic parameters can be determined [3].

Apparatus & Measurement Plan

The MgO crystal sample is placed in a microwave resonant cavity driven by a Varian Reflex Klystron operated in continuous-wave (CW) mode [9]. The Klystron Power Supply will be set such that the current does not exceed 40 mA. A primary electromagnet, powered by a linear ramp supply, will generate a static magnetic field (~0 - 0.7 T) monitored using a Gaussmeter probe near the cavity. An Agilent Function Generator will produce a sinusoidal modulation signal (~kHz), which is amplified by a Bryston Audio Amplifier and applied to a low-impedance modulation coil, generating a weak secondary magnetic field [10]. A DSP Lock-in Amplifier will be used to isolate the in-phase and out-of-phase components of the cavity output near the modulating frequency [11].

In the first week, we will calibrate the Gaussmeter by using DPPH. A large modulation amplitude will be applied, and the audio wave (X) and cavity output (Y) will be fed into the digital oscilloscope (DSO). The primary magnetic field will be swept until the modulation window spans the resonant condition, $H_0 = \frac{h\nu}{g\mu_B}$, at which point a closed loop absorption trace appears on the DSO [12]. Using this, we can calculate the true field at resonance for the known g-value of DPPH and determine the calibration factor for the Gauss probes. In the second week, the audio waveform will be used as the lock-in frequency for the cavity output. We will perform preliminary sweeps by turning on the linear voltage ramp and data collection simultaneously, matching the ramp rate with the lock-in amplifier time constant and choosing an appropriate sampling time until a clean ESR spectrum is observed. In the last week, we will perform multiple sweeps at different microwave orientations to identify the Fe^{3+}, Cr^{3+}, and Mn^{2+} hyperfine resonances.

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